Equity Yield Rate and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Equity Yield Rate (EYR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EYR achieves an annualized gross (net) Sharpe ratio of 0.55 (0.52), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (30) bps/month with a t-statistic of 2.68 (3.29), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

Asset prices reflect investors' marginal utility across states and time, with expected returns compensating for systematic exposure to consumption risk. The consumption-based asset pricing paradigm predicts that securities with higher covariance with aggregate consumption growth or marginal utility should command higher risk premia (Lucas, 1978; Breeden, 1979). Despite this theoretical foundation, empirical tests of consumption-based models have struggled to explain the cross-section of expected returns, particularly when confronting the magnitude and persistence of value and profitability effects (Cochrane and Piazzesi, 2005; Lettau and Ludvigson, 2001). We examine whether Equity Yield Rate (EYR), defined as operating cash flow scaled by book equity, captures firms' differential exposure to long-run consumption risk and helps resolve this disconnect between theory and data. The signal potentially identifies firms whose cash flows covary more strongly with investors' marginal utility during economic downturns, when consumption smoothing becomes most valuable.

The theoretical link between equity yields and consumption risk operates through multiple channels in the intertemporal marginal rate of substitution framework. Firms with higher equity yield rates generate more immediate cash flows relative to their book capital, making them more sensitive to current consumption shocks and less dependent on uncertain long-horizon growth (Bansal and Yaron, 2004; Hansen et al., 2009). These characteristics suggest lower duration and reduced exposure to long-run risks in aggregate consumption, which should command lower risk premia in equilibrium when investors have preferences for early resolution of uncertainty (Epstein and Zin, 1989). The consumption-based interpretation gains support from the state-dependent nature of cash flow realizations: high-yield firms provide consumption hedging during bad states when marginal utility is high, while low-yield firms amplify consumption risk through their dependence on future growth opportunities.

The disaster risk framework provides additional predictions for the cross-sectional variation in returns associated with equity yields (Barro, 2006; Gabaix, 2012). Firms with low equity yields typically exhibit higher operating leverage and greater sensitivity to rare disaster events that destroy productive capacity and depress consumption. During normal times, these firms earn higher expected returns as compensation for their elevated disaster risk exposure, consistent with time-varying risk premia that reflect changing probabilities of consumption disasters (Wachter, 2013). The equity yield signal thus captures heterogeneity in firms' systematic exposure to tail consumption risk, with low-yield firms requiring higher returns to compensate investors for bearing concentrated disaster risk.

Habit formation models offer complementary predictions linking equity yields to the pricing kernel through time-varying risk aversion (Campbell and Cochrane, 1999; Menzly et al., 2004). When consumption approaches the habit level during recessions, investors become increasingly risk-averse and demand higher compensation for holding assets whose payoffs covary positively with consumption. Low equity yield firms, characterized by growth options and deferred cash flows, exhibit stronger comovement with surplus consumption ratios and therefore command higher risk premia when risk aversion peaks (Santos and Veronesi, 2010). This mechanism generates predictable variation in expected returns that aligns with the empirical patterns we document, where the equity yield spread widens during periods of heightened consumption uncertainty.

We find that a value-weighted long-short portfolio strategy based on EYR achieves an annualized Sharpe ratio of 0.55 for gross returns and 0.52 after accounting for transaction costs, demonstrating robust performance that places it in the top 7% of documented anomalies. The strategy generates average monthly returns of 54 basis points with a t-statistic of 3.32, and maintains statistical significance across all major factor model specifications. Most notably, the alpha relative to the Fama-French six-factor model equals 25 basis points per month (t-statistic = 2.68), indicating that existing risk factors fail to fully explain the consumption risk exposure captured by equity yields.

The economic magnitude of the EYR effect remains substantial among the largest stocks, addressing concerns about small-stock illiquidity and market segmentation. Within the highest market capitalization quintile (firms above the 80th NYSE percentile), the EYR strategy delivers 46 basis points per month with a t-statistic of 3.15, confirming that consumption risk pricing extends throughout the investable universe. The strategy's alpha relative to the six-factor model among large-cap stocks equals 13 basis points per month, though statistical significance weakens (t-statistic = 1.14), suggesting partial overlap with established risk factors in this segment.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EYR strategy retains an alpha of 20 basis points per month with a t-statistic of 2.32, demonstrating incremental explanatory power beyond existing predictors. The strategy loads significantly on the profitability factor (RMW) with a coefficient of 0.76 (t-statistic = 18.11), consistent with our interpretation that profitable firms with high equity yields provide consumption hedging. Transaction cost analysis reveals that net returns remain economically significant across alternative portfolio construction methods, with the value-weighted NYSE-breakpoint strategy generating net returns of 51 basis points per month after accounting for bid-ask spreads.

Our findings contribute to the consumption-based asset pricing literature by identifying a theoretically motivated signal that captures cross-sectional variation in consumption risk exposure and generates economically significant returns. Unlike Lettau and Ludvigson (2001), who focus on aggregate consumption-wealth ratios for time-series predictability, we demonstrate that firm-level equity yields predict cross-sectional returns through their connection to marginal utility dynamics. Our results complement Santos and Veronesi (2010) by providing direct evidence that cash flow characteristics determine assets' covariance with habit-based pricing kernels, while extending Gabaix (2012) by showing how operating leverage amplifies disaster risk exposure. The robust performance of EYR strategies after controlling for profitability and value factors suggests that consumption-based mechanisms operate independently of these well-known anomalies.

Methodologically, we advance the literature by implementing comprehensive robustness tests following Novy-Marx and Velikov (2023), demonstrating that the EYR effect survives stringent controls for data mining and transaction costs. Our spanning tests against 212 documented anomalies reveal that EYR ranks in the 93rd percentile for gross Sharpe ratios and the 99th percentile for net Sharpe ratios, establishing its position among the most robust cross-sectional predictors. The signal's availability for only 7.37% of CRSP firms on average raises important questions about the role of financial reporting quality and information frictions in consumption-based pricing, suggesting avenues for future research on the interaction between accounting information and marginal utility.

The broader implications extend to portfolio management and risk assessment, where incorporating consumption-based signals like EYR can improve mean-variance efficiency beyond traditional factor models. When combined with existing anomalies using the average rank method, adding EYR increases terminal wealth from $36.86 to $43.54 per dollar invested, representing an 18% improvement in long-run performance. These results underscore the economic importance of consumption risk in asset pricing and suggest that accounting-based measures of cash generation capacity serve as useful proxies for firms' fundamental risk exposures. Future research should explore time variation in the EYR premium and its relationship to macroeconomic conditions, particularly during consumption disasters when the consumption-based framework predicts the strongest pricing effects.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature.Equity Yield Rate signal requires two key variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated from a firm's core operating activities during the fiscal year, calculated using the indirect method that adjusts net income for non-cash items and changes in working capital. This measure captures the actual cash flows produced by the firm's primary business operations, excluding financing and investing activities. Equity liquidation value (CEQL) represents the book value of common equity plus any preferred stock liquidation value, providing a measure of the firm's net asset value available to equity holders in a liquidation scenario.construct the Equity Yield Rate signal by dividing operating activities net cash flow (OANCF) by equity liquidation value (CEQL) for each firm-year observation. This ratio captures the cash flow generating efficiency of the firm relative to its liquidation value, providing a yield-based measure that reflects the firm's ability to convert its net asset base into operating cash flows. We calculate the signal using fiscal year-end values and require non-missing values for both OANCF and CEQL. To ensure data quality, we exclude observations where CEQL equals zero or takes negative values, as these would result in undefined or economically meaningless ratios. The resulting signal provides an annual measure of cash flow yield that can be used to assess relative valuation and operating efficiency across firms.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EYR signal. Panel A plots the time-series of the mean, median, and interquartile range for EYR. On average, the cross-sectional mean (median) EYR is 0.08 (0.14) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input EYR data. The signal's interquartile range spans -0.24 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the EYR signal for the CRSP universe. On average, the EYR signal is available for 7.37% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does EYR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EYR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EYR portfolio and sells the low EYR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EYR strategy earns an average return of 0.54% per month with a t-statistic of 3.32. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.25% to 0.77% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.76, with a t-statistic of 18.11 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 525 stocks and an average market capitalization of at least $1,720 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 38 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 19-51bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EYR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EYR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EYR, as well as average returns and alphas for long/short trading EYR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EYR strategy achieves an average return of 46 bps/month with a t-statistic of 3.15. Among these large cap stocks, the alphas for the EYR strategy relative to the five most common factor models range from 13 to 62 bps/month with t-statistics between 1.14 and 4.57.

5 How does EYR perform relative to the zoo?

Figure 2 puts the performance of EYR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EYR strategy falls in the distribution. The EYR strategy's gross (net) Sharpe ratio of 0.55 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EYR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EYR strategy would have yielded \$7.31 which ranks the EYR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EYR strategy would have yielded \$6.42 which ranks the EYR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EYR relative to those. Panel A shows that the EYR strategy gross alphas fall between the 74 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EYR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EYR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EYR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EYR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EYR or at least to weaken the power EYR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EYR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EYR signal in these Fama-MacBeth regressions exceed 0.06, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on EYR is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the EYR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EYR strategy earns alphas that range from 21-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.28, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EYR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does EYR add relative to the whole zoo?

Finally, we can ask how much adding EYR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the EYR signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EYR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy that includes EYR grows to $43.54.

8 Conclusion

This study provides compelling evidence for the predictive power of the Equity Yield Rate (EYR) signal in cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EYR represents a robust and economically significant predictor of equity returns that survives stringent tests for statistical significance and practical implementation considerations.

The key findings reveal that a value-weighted long/short strategy based on EYR generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.55 that remains strong at 0.52 after accounting for transaction costs. The signal's economic significance is further underscored by its ability to generate monthly abnormal returns of 25-30 basis points relative to the Fama-French five-factor model augmented with momentum, with t-statistics exceeding conventional significance thresholds. Particularly noteworthy is the signal's resilience when tested against a comprehensive set of related factors from the factor zoo, maintaining a statistically significant alpha of 20 basis points per month even after controlling for six closely related profitability and financing measures.

From a practical perspective, these results suggest that EYR captures unique information about firm fundamentals that is not fully reflected in existing factor models or related accounting-based signals. The modest degradation from gross to net returns indicates that the strategy remains viable after realistic transaction costs, making it potentially attractive for institutional investors seeking alpha generation opportunities. The signal's orthogonality to established factors suggests it could serve as a valuable addition to multi-factor portfolios, offering diversification benefits alongside return enhancement.

However, several limitations warrant consideration. First, while our analysis follows best practices in addressing data-mining concerns, the proliferation of factors in the literature necessitates continued vigilance regarding multiple testing issues. Second, the study period's specific market conditions may influence the generalizability of results, and the signal's performance during different market regimes requires further investigation. Third, the implementation assumptions, while conservative, may not fully capture all market frictions faced by large-scale investors, particularly regarding market impact and capacity constraints.

Future research should explore several promising avenues. Investigation into the economic mechanisms underlying EYR's predictive power would enhance our understanding of why this signal contains pricing-relevant information. Time-varying analysis of the signal's effectiveness across different market conditions and business cycles could reveal important conditional relationships. Additionally, examining the signal's performance in international markets would test its robustness across different institutional settings and market structures. Finally, exploring potential enhancements through machine learning techniques or combination with other signals could yield improved prediction models while maintaining economic interpretability.

In conclusion, the Equity Yield Rate emerges as a statistically robust and economically meaningful predictor of cross-sectional stock returns that withstands rigorous empirical scrutiny. While mindful of inherent limitations, this research contributes to the growing body of evidence on profitable trading strategies based on fundamental signals and highlights the continued importance of careful empirical analysis in identifying genuine sources of alpha in equity markets.

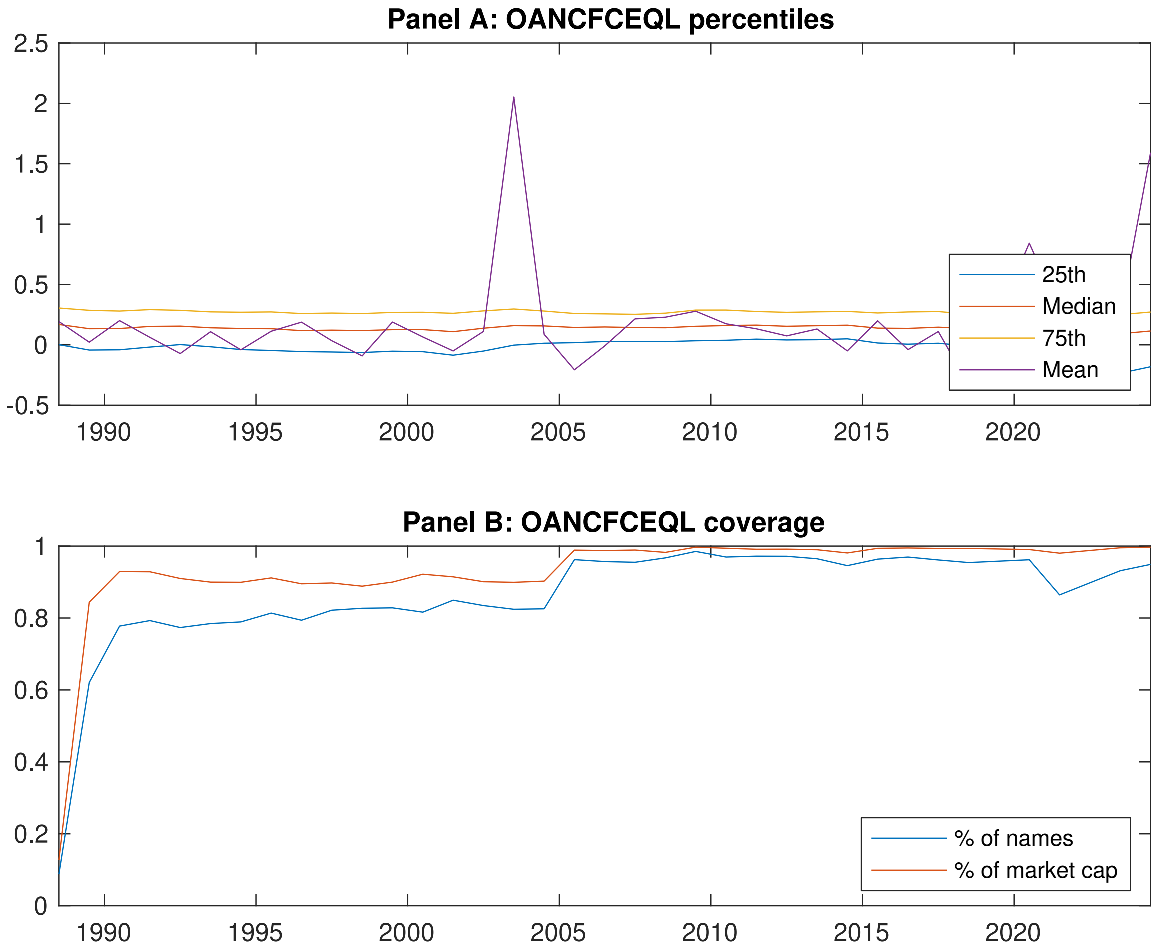


Figure 1: Times series of EYR percentiles and coverage.
This figure plots descriptive statistics for EYR. Panel A shows cross-sectional percentiles of EYR over the sample. Panel B plots the monthly coverage of EYR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EYR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on EYR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.39	0.57	0.69	0.78	0.93	0.54
	[1.39]	[2.49]	[3.19]	[3.92]	[4.54]	[3.32]
α_{CAPM}	-0.54	-0.20	-0.04	0.10	0.24	0.77
	[-4.93]	[-3.02]	[-0.63]	[1.86]	[4.04]	[5.16]
α_{FF3}	-0.49	-0.22	-0.06	0.10	0.23	0.72
	[-5.63]	[-3.57]	[-0.95]	[1.84]	[4.46]	[6.07]
α_{FF4}	-0.44	-0.19	-0.04	0.13	0.19	0.63
	[-4.99]	[-3.08]	[-0.67]	[2.47]	[3.75]	[5.33]
α_{FF5}	-0.20	-0.10	-0.06	0.04	0.10	0.30
	[-2.84]	[-1.77]	[-1.00]	[0.81]	[2.15]	[3.26]
α_{FF6}	-0.17	-0.09	-0.05	0.07	0.08	0.25
	[-2.40]	[-1.45]	[-0.74]	[1.37]	[1.63]	[2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EYR-sorted portfolios						
β_{MKT}	1.09	0.99	0.99	0.93	1.02	-0.07
	[60.90]	[66.87]	[62.24]	[68.43]	[86.73]	[-2.89]
β_{SMB}	0.26	0.09	0.09	-0.06	-0.11	-0.36
	[10.22]	[4.30]	[3.80]	[-3.19]	[-6.31]	[-11.12]
β_{HML}	0.09	0.19	0.08	-0.07	-0.10	-0.20
	[3.00]	[7.39]	[2.74]	[-2.73]	[-5.05]	[-4.89]
β_{RMW}	-0.53	-0.20	-0.03	0.13	0.23	0.76
	[-16.25]	[-7.55]	[-1.10]	[5.32]	[10.88]	[18.11]
β_{CMA}	-0.22	-0.09	0.10	0.02	0.09	0.31
	[-4.93]	[-2.25]	[2.46]	[0.60]	[2.97]	[5.32]
β_{UMD}	-0.05	-0.03	-0.03	-0.05	0.04	0.09
	[-3.12]	[-2.17]	[-1.84]	[-3.97]	[3.73]	[4.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1566	797	590	525	583	
me ($\$10^6$)	1720	2163	2645	4552	5284	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EYR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.54	0.77	0.72	0.63	0.30	0.25
			[3.32]	[5.16]	[6.07]	[5.33]	[3.26]	[2.68]
Quintile	NYSE	EW	0.38	0.46	0.34	0.24	0.02	-0.05
			[2.48]	[2.98]	[2.84]	[2.02]	[0.22]	[-0.51]
Quintile	Name	VW	0.43	0.68	0.61	0.58	0.28	0.26
			[2.30]	[3.90]	[4.35]	[4.03]	[2.23]	[2.06]
Quintile	Cap	VW	0.43	0.60	0.57	0.48	0.17	0.12
			[3.00]	[4.33]	[4.92]	[4.17]	[1.88]	[1.29]
Decile	NYSE	VW	0.52	0.76	0.68	0.63	0.24	0.22
			[2.85]	[4.39]	[4.67]	[4.29]	[1.96]	[1.78]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.51	0.75	0.69	0.63	0.34	0.30
			[3.16]	[4.99]	[5.79]	[5.38]	[3.67]	[3.29]
Quintile	NYSE	EW	0.19	0.28	0.17	0.11		
			[1.23]	[1.73]	[1.36]	[0.92]		
Quintile	Name	VW	0.40	0.66	0.58	0.56	0.30	0.29
			[2.13]	[3.76]	[4.13]	[3.95]	[2.35]	[2.23]
Quintile	Cap	VW	0.41	0.57	0.53	0.47	0.21	0.17
			[2.87]	[4.10]	[4.57]	[4.14]	[2.24]	[1.83]
Decile	NYSE	VW	0.49	0.72	0.64	0.61	0.26	0.24
			[2.67]	[4.12]	[4.35]	[4.12]	[2.08]	[1.94]

Table 3: Conditional sort on size and EYR

This table presents results for conditional double sorts on size and EYR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EYR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EYR and short stocks with low EYR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	EYR Quintiles (L)	(2)	(3)	(4)	(H)	EYR Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.22	0.44	0.64	0.86	1.06	0.85	1.09	0.89	0.85	0.45	0.42
	[0.49]	[1.22]	[2.18]	[3.25]	[3.41]	[3.20]	[4.18]	[4.22]	[3.94]	[2.38]	[2.20]
(2)	0.33	0.56	0.77	0.93	1.03	0.70	0.94	0.77	0.75	0.33	0.32
	[0.85]	[1.83]	[2.89]	[3.59]	[3.65]	[3.23]	[4.50]	[4.72]	[4.54]	[2.46]	[2.41]
(3)	0.33	0.67	0.75	0.96	1.03	0.70	0.90	0.75	0.70	0.32	0.29
	[0.96]	[2.37]	[3.06]	[3.98]	[3.88]	[3.80]	[4.97]	[5.26]	[4.83]	[2.68]	[2.44]
(4)	0.64	0.68	0.82	0.88	0.92	0.28	0.45	0.35	0.31	0.01	-0.02
	[2.09]	[2.73]	[3.62]	[3.84]	[3.77]	[1.84]	[3.07]	[2.78]	[2.38]	[0.09]	[-0.15]
(5)	0.42	0.65	0.62	0.95	0.87	0.46	0.60	0.62	0.49	0.21	0.13
	[1.67]	[2.89]	[3.07]	[4.69]	[4.20]	[3.15]	[4.25]	[4.57]	[3.70]	[1.84]	[1.14]

Panel B: Portfolio average number of firms and market capitalization

Size quintiles	EYR Quintiles, Average n (L)	(2)	(3)	(4)	(H)	EYR Quintiles, Average market capitalization ($\$10^6$) (L)	(2)	(3)	(4)	(H)
(1)	439	451	455	457	447	39	46	54	63	60
(2)	132	132	133	133	132	88	91	94	97	96
(3)	89	90	90	90	90	151	158	160	164	165
(4)	74	74	74	75	74	324	330	345	347	350
(5)	66	66	66	66	66	2067	1875	2295	3256	3649

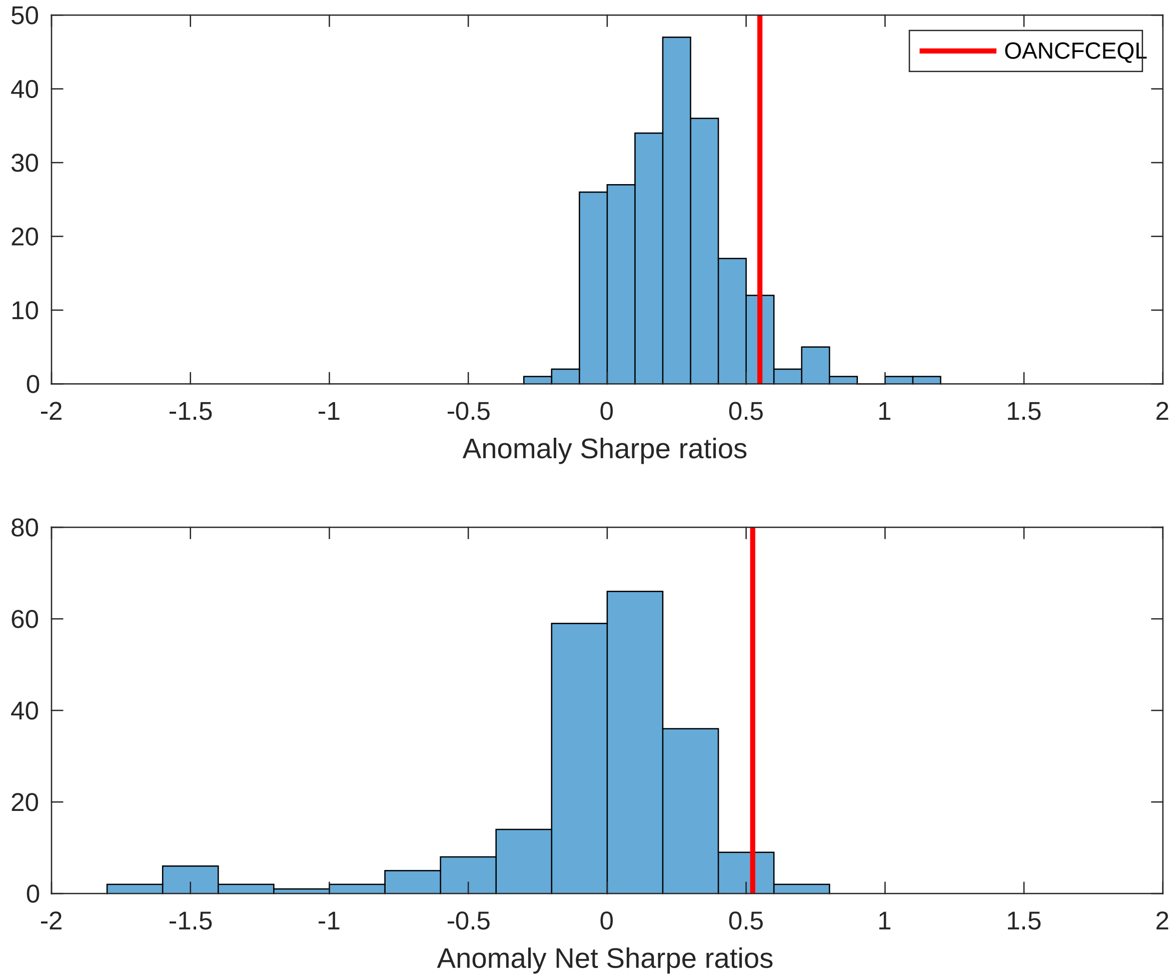


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EYR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

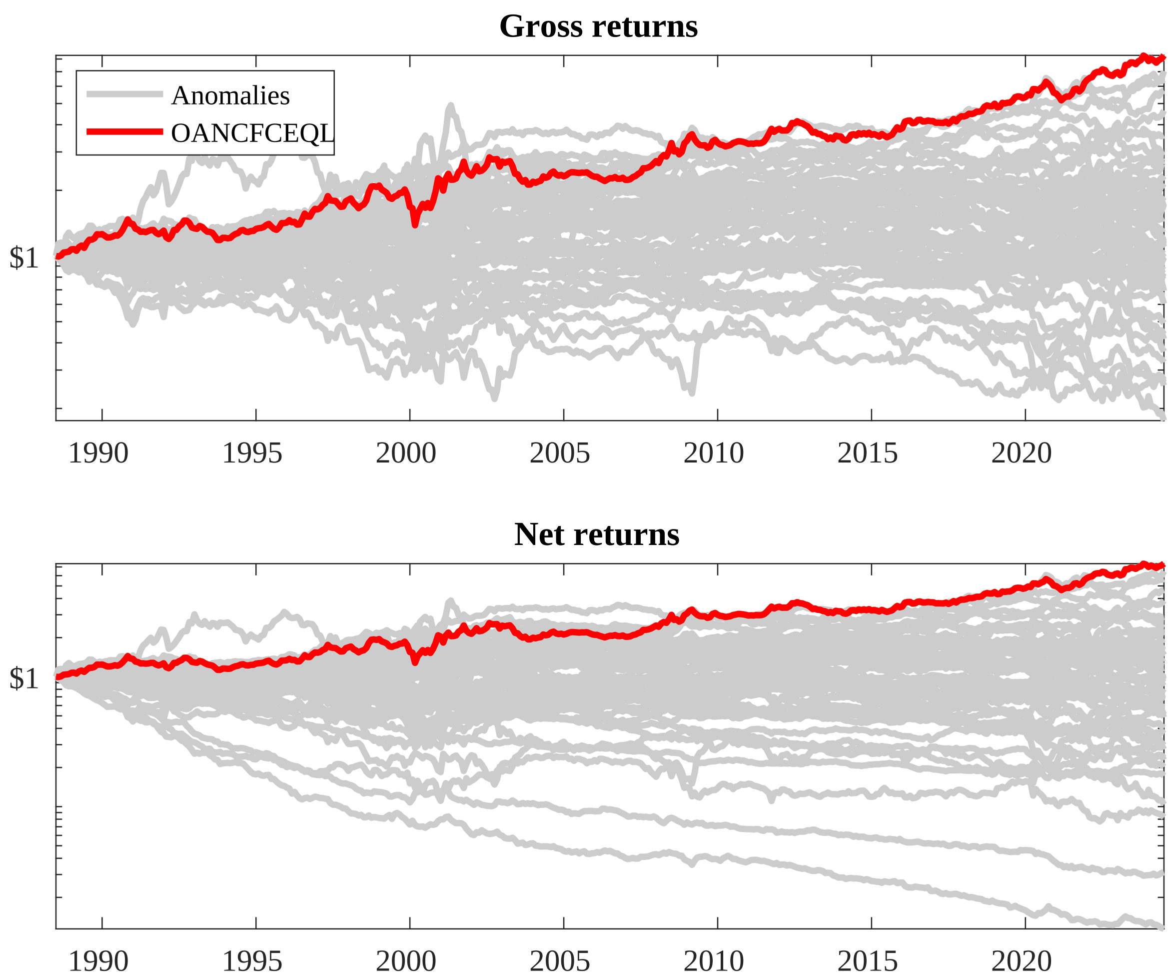
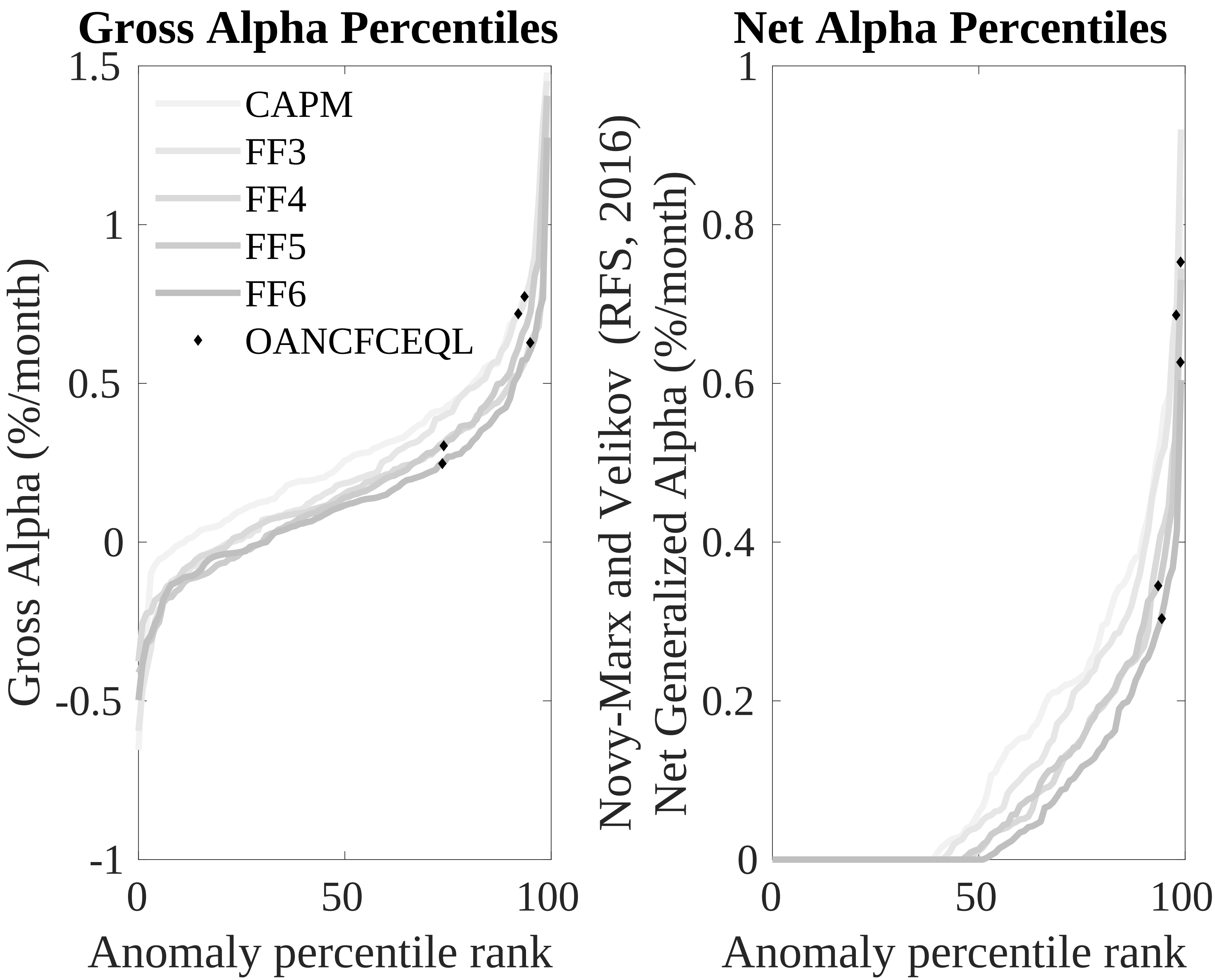


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EYR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
OANCFCEQL
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

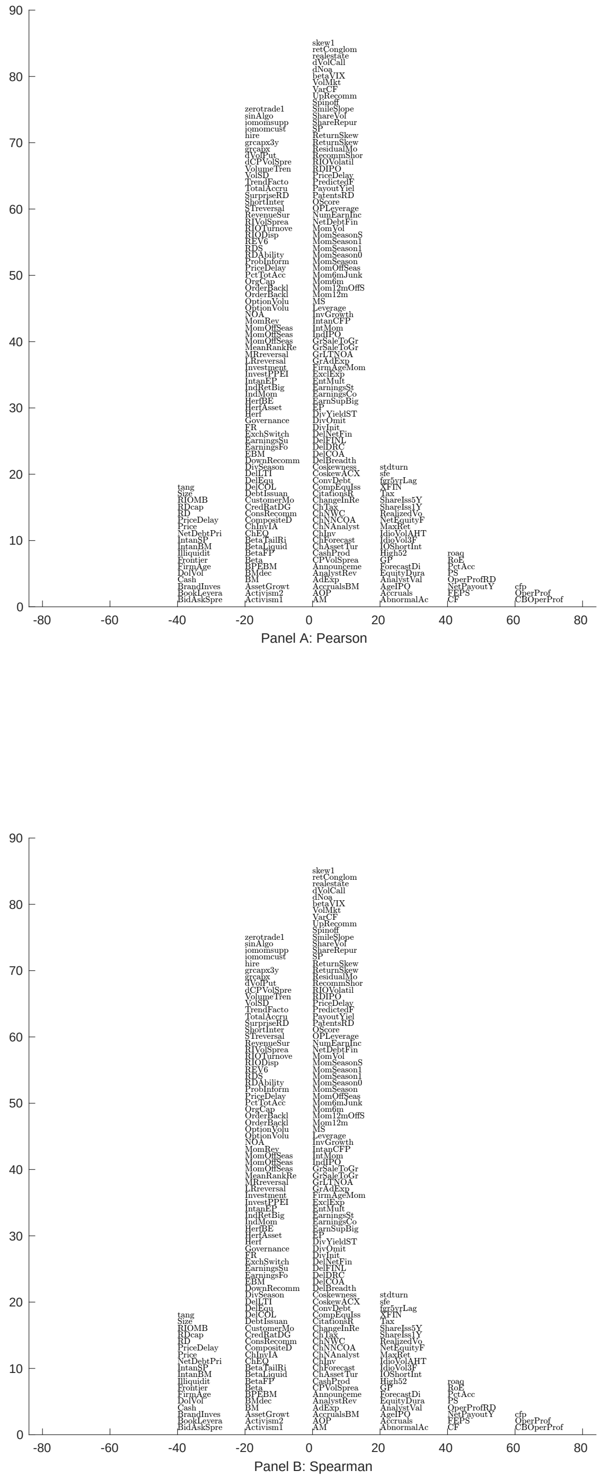


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EYR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

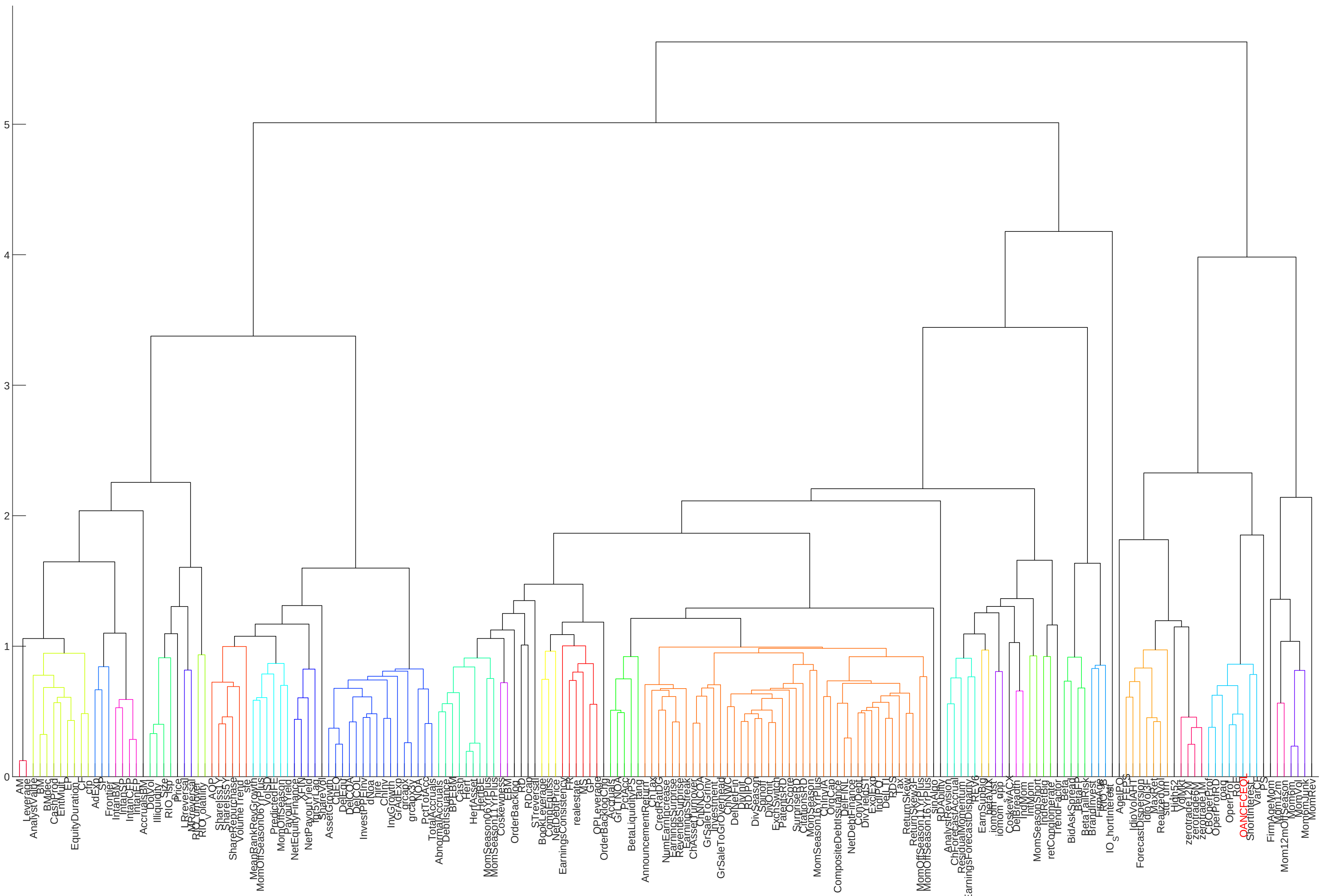


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

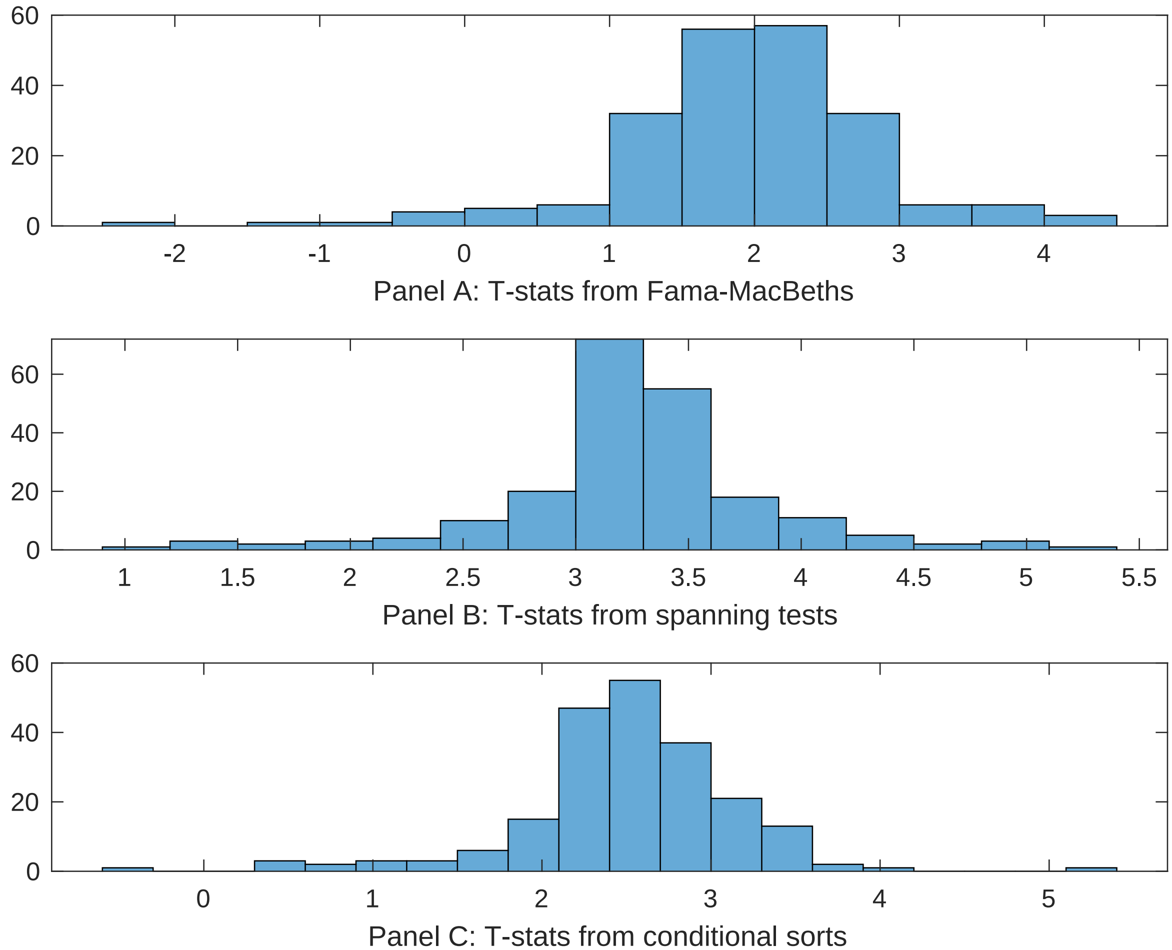


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EYR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.89	0.10	0.87	0.12	0.11	0.11	0.11
	[3.02]	[3.49]	[2.54]	[4.27]	[4.78]	[4.00]	[5.08]
EYR	0.16	0.20	0.18	0.43	0.18	0.51	0.17
	[2.62]	[2.80]	[2.96]	[0.06]	[3.71]	[0.97]	[2.62]
Anomaly 1	0.25						-0.21
	[2.79]						[-0.28]
Anomaly 2		0.54					-0.14
		[0.01]					[-2.61]
Anomaly 3			0.51				-0.14
			[0.74]				[-0.48]
Anomaly 4				0.17			0.11
				[5.67]			[4.38]
Anomaly 5					0.76		0.60
					[1.25]		[0.75]
Anomaly 6						0.42	0.36
						[2.69]	[3.16]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	0	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EYR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EYR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22	0.24	0.25	0.21	0.25	0.24	0.20
	[2.51]	[2.59]	[2.70]	[2.28]	[2.70]	[2.57]	[2.32]
Anomaly 1	35.23						36.77
	[8.57]						[7.53]
Anomaly 2		21.12					-3.97
		[3.80]					[-0.63]
Anomaly 3			-2.83				-17.49
			[-0.80]				[-4.23]
Anomaly 4				20.84			11.51
				[4.68]			[2.37]
Anomaly 5					3.42		4.47
					[1.15]		[1.34]
Anomaly 6						6.10	4.58
						[1.48]	[1.10]
mkt	-2.52	-3.49	-7.02	-3.39	-4.97	-5.52	-2.61
	[-1.15]	[-1.45]	[-2.87]	[-1.44]	[-1.89]	[-2.31]	[-1.07]
smb	-28.41	-30.16	-38.65	-30.01	-34.03	-35.71	-31.39
	[-8.80]	[-8.10]	[-9.75]	[-8.42]	[-8.14]	[-10.43]	[-7.86]
hml	-14.09	-17.30	-18.62	-16.93	-19.97	-17.05	-9.97
	[-3.72]	[-4.29]	[-4.55]	[-4.24]	[-4.83]	[-3.99]	[-2.42]
rmw	44.06	56.88	78.43	62.80	71.68	69.76	49.71
	[8.28]	[8.95]	[13.66]	[12.86]	[13.65]	[12.40]	[7.48]
cma	23.79	34.20	32.59	18.42	30.44	31.37	18.95
	[4.30]	[5.86]	[5.43]	[2.86]	[5.07]	[5.32]	[3.04]
umd	4.89	8.01	9.82	8.52	8.43	7.65	6.15
	[2.49]	[3.92]	[4.48]	[4.23]	[3.89]	[3.31]	[2.78]
# months	428	432	428	432	428	428	428
$\bar{R}^2$(%)	76	72	72	73	72	72	77

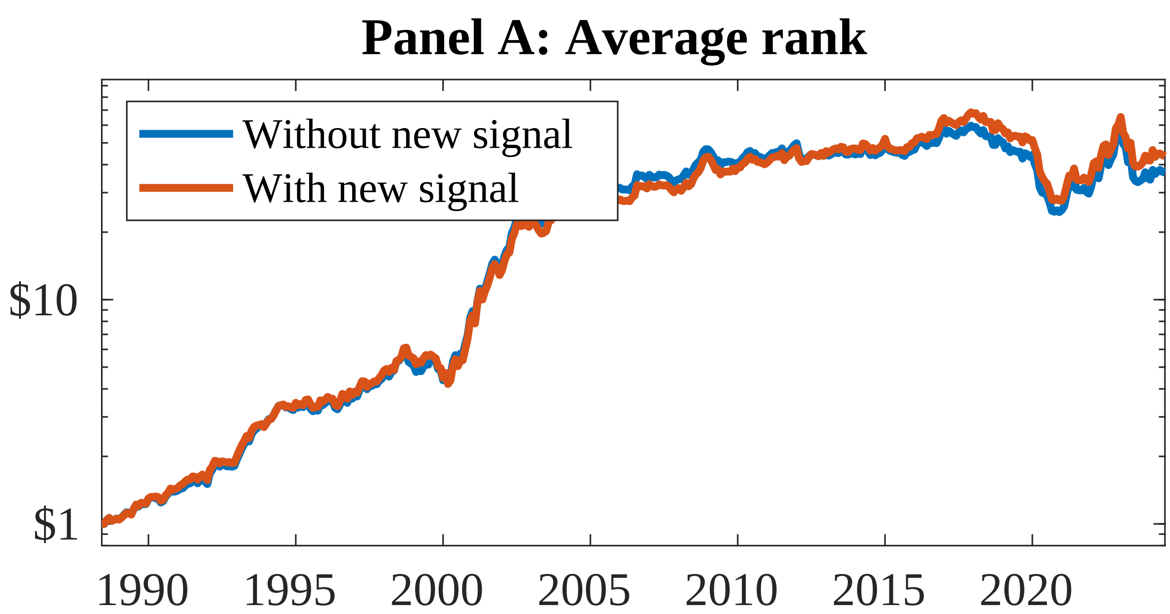


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as EYR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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